What is information

Information n. Data ~~connected~~ communicated or received that resolves uncertainity about a particular fact or circumstance

Quantify Information

Given discrete random variable X

N possible value x_1 x_2 x_3 x_4

Associated probabilities p_1 p_2 p_3 p_4

Information received when learning that choice was x_i

$$I(x_i) = \log_2\left(\frac{1}{p_i}\right) \text{ bits}$$

I = measured in bits

$$x_i \propto \frac{1}{p_i}$$

Information convey by data

When data cannot resolve all the uncertainity

Common case Suppose you're faced with N equally probable choices and you received data that narrow down to m choices

data probability = $m(1/N)$

Amount of Information received = $I(\text{data}) = \log_2\left(\frac{1}{m(1/N)}\right) = \log_2\left(\frac{N}{m}\right)$ bits

Entropy

In Information theory, the entropy $H(X)$ is the average amount of information contained in each piece of data received about the value of X.

$$H(X) = E(I(X)) = \sum_{i=1}^{N} p_i \cdot \log_2\left(\frac{1}{p_i}\right)$$

Encoding

An encoding is an unambiguous mapping b/w bits string and the set of possible data.

A	B	C	D
00	01	10	11
0	1	10	11

Encoding for "ABBA"

00 01 01 00

0 1 1 0

ABBA?

ABC?

ADA?

Encoding as Binary tree

Encoding

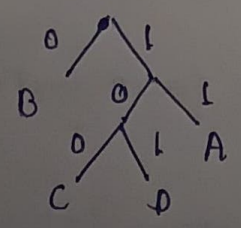
B → 0

A → 11

C → 100

D → 101

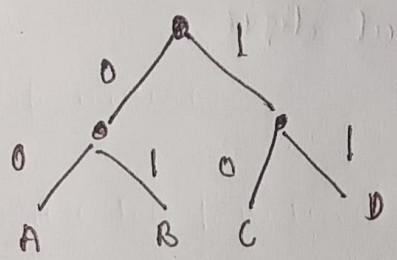
Binary tree



Fixed length ~~Cod~~ Encoding (FLC)

If all choices are equally likely (or we have ~~to~~ no reason to expect otherwise), then FLC is used.

- Such code will use at least enough bits to represent the information content.



* All leaves have same length

$$\sum_{i=1}^{N} \left(\frac{1}{N}\right) \log_2\left(\frac{1}{1/N}\right) = \log_2 N$$

* fixed length often a little inefficient

Ex 4 bits BCD digits. $\log_2(10) = 3.322$

Encoding the numbers

Easy to encode the integers as a sequence of bits.

Each bits is assigned a weight ordered from R to L.

Weights are increasing power of 2.

$$v = \sum_{i=0}^{N-1} 2^i b_i$$

smallest no : 0
(when all binary digits are zero)

Largest no : $2^N - 1$
(when all binary digits are 1)

Hexadecimal Notation

- Higher radix notation
- Choosing the radix so that it is simple to recover the original bit string
- Hexadecimal → where each group of 4 adjacent bits are represented as a single hexadecimal digit

0b 0111110100000 = 0x7D0

0x → special prefix to represent hex.

Encoding signed integer

N bits

- Range $-(2^{N-1}-1)$ to $2^{N-1}-1$

In decimal	In Binary
'+' → positive	'0' → ~~neg~~ positive
'-' → negative	'1' → negative

Two's complement encoding

In this, the high order bit of the N bit representation has negative weight.

N bits

-2^{N-1} 2^{N-2} - - - - - - 2^3 2^2 2^1 2^0

- Negative numbers has '1' in the highest order bit

Range -2^{N-1} to $2^{N-1}-1$

Variable length encoding

GOAL When encoding data we'd like to match the length of the encoding to the information content of data.

This means Higher probability → shorter encodings

Lower propbability → Longer encodings

such encoding is called variable length encoding

Ex

choice,	Pi	encoding
A	1/3	11
B	1/2	0
C	1/12	100
D	1/12	101

Expected length for this encoding

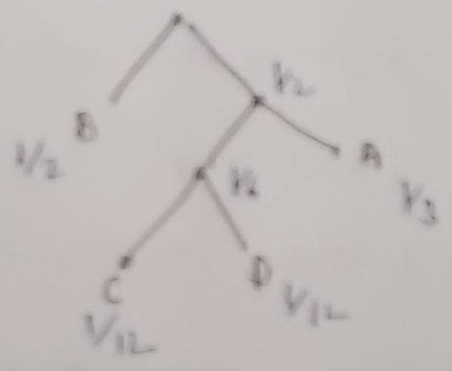
$$(2) \times 1/3 + (1)\, 1/2 + (3) \times (1/12) \wedge 2 = 1.667 \text{ bits}$$

For 1000 symbols

- with fixed length 2bit/symbols = 2000 bits
- with variable length code = 1667 bits
- Lower bound entropy = 1626 bits

Huffman Alogorithm

- Build a subtree using 2 symbols with lowest pi
- At each step choose two symbols / subtrees with lowest pi, combine to form new subtree
- Result optimal tree built from the bottom up

Example

A = 1/3 B = 1/2 C = 1/12 D = 1/12